vasic-digital tmux — v1.0.45 Release Notes

Revision: 3 **Last modified:** 2026-09-02T06:32:13Z **Version:** 1.0.45 / versionCode 46 **Released:** NOT YET — tag `tmux-1.0.45` is not cut. Intended 2026-09-01. **Hosts validated:** the full `verify.sh` sweep COMPLETED GREEN on this tree (`PASS=79 FAIL=0 SKIP=11`, 2026-09-02T06:09Z) — see "Validation" below for the result, the enumerated topology skips, and its honest boundary. §11.4.185 manual QA has NOT been performed and is not claimed (§11.4.6).

Summary

v1.0.45 is a **workable-items tracker-integrity** release. The tmux product surface is UNCHANGED from v1.0.44 — the same hardened `tmux 3.7b` binary, the same `tmx` wrapper, the same submodule pin (`e802909de06012a4df6209d55e86487c56223163`). Every change is in the §11.4.93 SQLite single-source-of-truth engine (`cmd/workable-items/`), the release gate (`scripts/verify.sh`), and the tracker corpus itself.

The cycle's defects share one shape: **a parser, a check, or a gate returning a confident quiet zero while enforcing nothing.** A heading regex that made a whole class of tracker block invisible to the SSoT; an identity audit that discarded a second ownership claim before it could be reported; two shipped governance gates that no runner ever invoked; and a test assertion that could not distinguish a skipped audit from a clean one. Fixing the first exposed six more defects in the sync engine that its narrowness had been masking.

This release is not finished. The independent review returned NO-GO and has not yet iterated to a clean GO (§11.4.134), four reviewer-authored mutations survived the author's test suite, and several proven root causes have fixes that are OWED rather than done. All of it is enumerated under "Known-open" below rather than implied complete.

Fixed

Parser / SSoT

- **Space-form headings were silently dropped from the SSoT.** `parser.go`'s `headingRE` required a period after the block ordinal (`^###\s+([A-Z])(\d+)\.\s+(.*)$`), so `### G5 NAME-001 — …` never parsed. Measured consequence: the block's body was absorbed into the PRECEDING item's `raw_body`, and its own row sat in the DB with a sentinel identity (category `Z` / `code_ordinal 0`), so nothing could locate it in the markdown. The corpus carries both forms — 23 of the `Fixed.md` block headings were space (measured at tag `tmux-1.0.44`; the corpus repair in this same release adds more) form — and the reconciler's own matcher `blockCodeRE` had ALWAYS accepted both, so the narrow parser was an internal inconsistency between two components of one tool, not a deliberate contract. Widened to `^###\s+([A-Z])(\d+)(?:\.\s+|\s+)(\S.*)$`.
- **Both regexes were TIGHTENED in the same change and are now changed together by construction.** `blockCodeRE` went from `^###\s+([A-Z])(\d+)[.\s]` to `^###\s+([A-Z])(\d+)(?:\.\s+|\s+)\S`, so the two agree on their accepted set. The separator is REQUIRED and must be followed by real title text, which correctly refuses `### A50X`, `### A1.x`, `### A12.5` (a decimal token would have yielded ordinal `12`, title `"5 …"`), `### A1.` and `### A1 ` (empty title — its `heading_hash` would be computed over `""`). Blast radius was measured BEFORE any sync: 90 loose-accepted corpus headings scanned, **0** dropped, control-needle proven (§11.4.201(7)(b)).
- The contract comment above `headingRE` is declared the §11.4.245 **specified oracle** for `heading_forms_test.go`: that test's accept/refuse table is derived from the prose contract, not from the regex, so a change to one that is not a change to the other FAILs.

The six defects the widening exposed (each fixed at root cause)

#	Defect	Fix
1	`UpsertItem` decided INSERT-vs-UPDATE on `heading_hash` alone, so a newly-parseable block died on `UNIQUE constraint failed:`	resolve an explicit `TMX-ID` to its existing row, UPDATE it, and REFRESH the hash so the next sync binds on the hash instead of repeating the miss

#	Defect	Fix
	`items.atm_id` (reproduced live for TMX-072)	
2	Two blocks for one item silently COLLAPSED into one row whose survivor depended on read order	fail-closed guard over the UNION of `ExplicitATM` and `HeadingHash` across every parsed block in BOTH files — it REFUSES rather than picking a winner
3	That guard persisted `document_sources` BEFORE running, so a REFUSED sync still mutated the git-tracked SSoT	raw text held in locals; both blobs persisted only after the guard passes (§11.4.252)
4	An identity rebind replaced a row's category / ordinal / title / body with NO history row and NO counter — `validate` reported 0 findings because the result was self-consistent	prior identity captured, `item_history` row written, counted in `SyncResult.RebindDetails`, printed by the CLI (§11.4.226)
5	`itemContentEqual` compared `Category` but not `CodeOrdinal` / `HeadingHash`, so an ordinal-only renumber took the "unchanged" path and the row kept a stale ordinal + hash PERMANENTLY	both fields compared (identity IS content); rows self-heal from either drift
6	`lookupByHeadingHash` matched on error TEXT and returned `("", nil)` for real DB errors — a failed query read as a hash MISS, which mints a duplicate row	`errors.Is(err, sql.ErrNoRows)`; real errors propagate

`add` / `db-to-md` seams

- **TMX-094's fix was structurally suppressed on the live corpus — an `add`-created row still rendered into NO tracker (found during release verification, 2026-09-01).** `sourcelessItems` derived a block code from an item's stored `code_ordinal`, but every `workable-items add` row carries the SENTINEL `code_ordinal = 0`, rendering as code `A0` — and `Fixed.md:2585` holds a genuine

`### A0. Initial migration from ATMOSphere project …` block declaring no `**TMX-ID:**`. `codeOwners["A0"][""]` therefore answered for EVERY add-created row in category A, suppressing exactly the rows TMX-094 exists to render. The codebase's own rule at `validate_identity.go:163` ("ordinal sentinel — asserts no block claim") was contradicted. Measured: TMX-099/TMX-100 sat in the DB with `current_location = Issues` rendering into neither tracker (`sourceless count=0` against the real corpus). Fixed by skipping the stored-ordinal check when `CodeOrdinal <= 0`; the `atmOrdinal` fallback is untouched. Post-fix: both render as `### A99.` / `### A100.`, `issues_parsed` 10 → 12, round-trip byte-identical, `validate` 0 findings. Guarded by a §11.4.115 polarity test (`RED_MODE=1` reproduces `count=0`) plus a §11.4.201(1) false-positive guard, proven load-bearing by a paired §1.1 mutation. The original per-signal tests missed it because their synthetic fixtures contained no <CAT>0 block — a fixture-realism gap (§11.4.245).

- **TMX-093 — `add` hashed the ATM id where the parser hashes the BLOCK CODE.** `add.go` passed `atm` into `computeHeadingHash`, while `parser.go` builds `code := category + ordinal` (`parser.go:186`) before hashing it (`parser.go:201`). An `add`-created row could not bind to its own block by hash and survived only via the `ExplicitATM` rebind fallback — which REPLACES the row and refreshes the hash, **masking** the split behind something that looked like normal operation. Fixed by hashing the block code the writer actually emits (`category + atmOrdinal(atm)`, mirroring `sync_db_to_md.go:195-201`).

- **TMX-095 (NEW) — `add --category` accepted values rendering a heading the parser can never read.** Measured 2026-09-01: `--category=AB` renders `### AB1.`, `--category=1` renders `### 11.`; neither matches `headingRE`. The row lands in the SSoT and the writer emits its block, but the parser cannot see it — the same invisible-block class, reached through the `add` seam. `AddItem` now refuses anything that is not exactly one letter `A-Z`, with an actionable message; `A`, `Z`, lowercase `a`, and the empty default MUST still be accepted (§11.4.201(1) — a guard that refuses the real corpus is a defect, not a safe default). Surfaced as an honest out-of-scope gap by the stream fixing TMX-093 and FILED rather than narrated (§11.4.197).

- **TMX-094 — `db-to-md` never rendered `add`-created rows.** `reconcileLocations` only MOVES blocks that already exist in a blob; it never APPENDS one for an item present in `items` but in NEITHER blob. Since `add` writes an `items` row and nothing to `document_sources`, an add-created row was a DB row no tracker document ever showed — a silent loss of visibility at the requirements-intake layer (§11.4.202 / §11.4.197). Fixed with `sourcelessItems()` + `appendRenderedItems()`, gated on THREE independent signals because a false positive here DUPLICATES a block into the tracker, which is strictly worse than the omission it repairs: (a) `items.raw_body` empty; (b) no `TMX-ID` line anywhere in either blob; (c) no plausibly-own `### <CAT><N>` block — a bare code match is NOT sufficient, because heading codes are not unique (measured: 93 code headings, 85 distinct — 8 codes appear twice), and the ordinal is checked BOTH as the stored `CodeOrdinal` and as `writeItemBlock`'s `atmOrdinal` fallback because the two legitimately disagree for legacy rows. Signals (b) and (c) are each individually insufficient: the id-line signal alone flags 62 of 95 items, **61 of them false**. Measured on the live corpus: **0** rows have an empty `raw_body`, so the pass appends nothing today and the live regeneration stays byte-identical.

Identity audit

- **The §11.4.54 audit DISCARDED a second ownership assertion in silence.** `markdownBlockOwners` was first-declarer-wins: `Fixed.md` carried 25 blocks declaring a `TMX-ID` while the owner map held 24 keys — `### A52` headed two blocks and one claim vanished, so the audit could not report a collision it had already thrown away. Duplicates are now RETURNED and raised as findings naming every claimant. Cross-FILE reuse of a code is deliberately NOT a duplicate — identity is `(location, category, ordinal)`, and the live corpus legitimately reuses 8 codes across the two trackers (§11.4.201(1)).

- **A code-less `###` heading did not reset the id-scan cursor**, so an id line could bind to the PRECEDING block. The reset is now keyed to ANY heading level (`anyHeadingRE`), matching the item parser's own greedy-bind guard — the two components had disagreed about what closes an id-scan window. Honest boundary (§11.4.6): measured by an awk walk of both trackers comparing the two reset rules, **ZERO** id lines change attribution. This closes a divergence and a live trap for the moment the missing id lines are backfilled; it does not repair a mis-attribution happening today.

- **The `### A52` block code was declared by BOTH TMX-062 and TMX-071** — a genuine §11.4.19 one-item-one-block violation with measured data loss. Both blocks live in `Fixed.md`: the space-form

`### A52 NO-SUDO-…` at line 2890 (TMX-062, itself a live specimen of the form this release made parseable) and the period-form `### A52. META-TEST-72-73-REGISTER-001` at line 3027 (TMX-071). The literal `### A52.` occurs exactly ONCE; it is the CODE that repeats, across the two forms — measured at HEAD `5191e82`, BEFORE the repair: `grep -c '^### A52\.'` = 1, `grep -cE '^### A52(\.| )'` = 2. Resolved by renumbering the TMX-071 block to `### A56.` (A56 was free; the highest in use was A55), with the sync's `ExplicitATM` path RECORDING the identity rebind. Post-renumber those commands measure `0` and `1`, and the owner map holds 25 keys. The superseded measurements are PINNED to their revision in the closure evidence rather than deleted, because the collision they document was real (§11.4.6).

- **TMX-078's stored identity named a block owned by a different ticket** — and no check could see it (landed in `ec25327`). The row stored `{category:A, code_ordinal:50, location:Fixed}` while `Fixed.md:2945` `### A50` declared `TMX-057`, and TMX-078's real block was `Issues.md:271` `### G5 SANITIZE-NAME-001`. `validate` reported 0 findings throughout because no check compared a row's STORED identity against the MARKDOWN block it names — a missing check and a clean corpus return the identical quiet zero. It is not catchable DB-internally either: a duplicate-triple scan misses it, because TMX-078's triple IS unique (the colliding item carries the ordinal-0 sentinel the narrow parser left behind). Only the markdown can settle who owns a block. Repaired (one row; §9.2 pre-op backup before the UPDATE; DB WAL-checkpointed `TRUNCATE` before staging per §11.4.95) and covered by the new cross-surface detector `ValidateBlockIdentity`, wired into `runValidate` behind `--issues` / `--fixed`, with four false-positive guards each measured against the live corpus.

- **Closing an item could move its block identity onto a block another item already held — `close.go` wrote the destination regardless.** A close is a BLOCK-IDENTITY MIGRATION, not only a status change: the identity triple `(current_location, category, code_ordinal)` is a pointer to ONE markdown block, and flipping `current_location` re-points it from `(Issues, cat, ord)` to `(Fixed, cat, ord)`. Block codes are CATEGORY-LOCAL and are NOT unique across the two trackers, so that destination can already be occupied — producing exactly the two-items-one-block state `ValidateBlockIdentity` exists to report. `CloseItem` now checks the destination BEFORE any write, so the refusal is side-effect free and the row is left exactly as it was (§11.4.252 fail-closed), and the error names the colliding holder plus the concrete recovery (renumber to a free code in `Issues.md`, re-sync, close again). Three false-positive guards mirror `validate_identity.go`'s, so the two components agree on what "claims a block" means (§11.4.201(1)): `code_ordinal <= 0` is the unknown-ordinal sentinel every `add` row carries and can never collide; an `Obsolete` closure legitimately SHARES its successor's block (§11.4.90 — measured live, `(Fixed, B, 3)` is held by both TMX-001 and TMX-054); and the item's own row is excluded, so an idempotent double-close is never refused by itself. The migration is also RECORDED on the closure history event (`block identity <CODE> migrated <from> -> Fixed`, appended to any operator-supplied reason rather than replacing it) — a rewrite with no recorded evidence is indistinguishable from a row that was always that way (§11.4.226).

Changed

- **Two shipped governance gates that enforced NOTHING are now wired into the release gate.** `scripts/testing/tracker_structural_integrity.sh` and `scripts/review/review_round_record_test.sh` live outside `run_all.sh`'s `scripts/tests/[0-9][0-9]_*.sh` glob, so grep proved **zero** invocations from any runner: both self-tested green by hand while gating zero (§11.4.205; §11.4.227(A) — a named gate must be implemented or a registered deferral, never silent debt). Found as a BLOCKING review finding and confirmed by re-measurement before acceptance. Both now execute from `scripts/verify.sh` as Layer-1 static gates `CM-TRACKER-STRUCTURAL-INTEGRITY` and `CM-REVIEW-ROUND-RECORD-WIRED`, each validated in BOTH polarities (golden-TRUE PASS on the healthy tree with resolved evidence; golden-FALSE FAIL with `rc=1` when the script is absent), each FAIL printing an actionable `reproduce:` line, and a FAIL of either ABORTS `setup.sh` before the runtime suite (the shared `TRACKGOV_FAIL` block, which now also carries the third gate below).
- **The three code-less ### headings are REPAIRED, and a THIRD Layer-1 gate now covers the class so it cannot silently return.** The tracker grammar never required a `### ` heading to carry a block code, so hand-added headings were absorbed into the preceding block's `raw_body` and were invisible to the §11.4.93 SSoT — one architectural cause reached by three unrelated triggers (a sub-note, the ATM→TMX prefix migration `89324dc`, a §12.10 status log). All three now carry a parser-readable block code and a `**TMX-ID:**` line: `Issues.md`'s `M24-ESCAPE-001` became `### B54 M24-

`ESCAPE-001 … — FIXED` (TMX-098), and `Fixed.md`'s two became `### D2.` (TMX-096) and `### D3.` (TMX-097) — measured with `grep -c 'TMX-096\|TMX-097\|TMX-098' Issues.md Fixed.md` → `Issues.md:0 Fixed.md:8`, all three now resident in the closed tracker. The class is enforced by `scripts/testing/heading_grammar_gate.sh`, wired into `scripts/verify.sh` as Layer-1 static gate `CM-HEADING-GRAMMAR` (definition `scripts/verify.sh:935`, invocation `:1050`, its FAIL reaching `exit 1` through the same `TRACKGOV_FAIL` block as the two gates above). Measured live: `Issues.md` headings=12 coded=12 code-less=0; `Fixed.md` headings=86 coded=86 code-less=0; `rc=0`. It carries a §11.4.201(7)(b) control needle — 3 synthetic headings through the SAME scanner return `seen=3 coded=2 code-less=1`, so its zero is proven not blind — and a §11.4.115(F) RED/GREEN pair: against `HEAD:Issues.md` + `HEAD:Fixed.md` (the pre-repair artifact) it FAILs naming all three offenders by file and line, and against the repaired corpus it PASSes. Its self-test `scripts/testing/heading_grammar_gate_test.sh` reports `PASS=9 FAIL=0`, two of the nine being §1.1 mutations (classifier weakened → gate MISSES the code-less heading, `rc=0`; counter neutered → same), proving the verdict depends on the assertion and the count rather than on incidental output. Companion guide: `docs/scripts/heading_grammar_gate.md` (§11.4.18).
- **Tracker corpus repair** (operator-approved 2026-09-01, both decisions recorded): removed four stale `### G1..G4` duplicates present as full blocks in BOTH trackers (TMX-072..075, already closed in `Fixed.md`); resolved the `A50` double-claim by renumbering `Fixed.md`'s block to `### A55.` with `**TMX-ID:** TMX-078`, then retiring the duplicate `Issues.md ### G5`. The emptied `## G.` section was **ANNOTATED, not deleted**. Cross-tracker duplicates **5 → 0**, enforced at the sync seam.
- **`constitution` submodule pointer `b9096ac → f5876a3`** — a codegraph ledger append; no governance text changed.
- **`README.md`** gains a doc-link row for the two newly-wired gates and their companion guides, so both are reachable from the canonical entry point (§11.4.212).

Tests

- **New Go test files:** `heading_forms_test.go` (the specified-oracle accept/refuse table plus a `headingRE`/`blockCodeRE` agreement test), `duplicate_block_test.go`, `upsert_identity_test.go`, `guard_coverage_test.go`, `validate_identity_test.go`, `validate_identity_duplicate_test.go`, `add_hash_convention_test.go`, `add_category_validation_test.go`, `db_to_md_add_rows_test.go`, `sourceless_signals_test.go`; extended `parser_greedybind_test.go` and `reconcile_identity_test.go`.

- **Test 51 gate hardening.** T3 asserted only that the output contains `0 findings` — but a SKIPPED audit prints `0 findings` too, so the gate would have PASSed on a check that never ran (§11.4.201 false-null). T3 now refuses skip-shaped output explicitly, and was proven load-bearing by mutation: hide the tracker → FAIL; restore → no false fire.
- **Mutation ledger (§1.1): 19 guards pinned**, each mutation verified to LAND in the file before being run; nine of the 19 were authored by the independent reviewer rather than by the author. Recorded in `docs/qa/2026-09-01-heading-regex-widening/closure-evidence.md`.
- **Blind tests caught by RUNNING the mutations rather than trusting the green.** Two of the author's own: the rebind test's assertions sat behind an inverted condition that skipped them every run; the precedence test asserted an outcome identical under both orderings. Two of `ec25327`'s controls, for the same reason: a helper left `HeadingHash` empty so every fixture `put()` overwrote the previous row (the 2-item obsolete fixture collapsed to 1), and the ordinal-0 control asserted only `findings==0`, which the detector satisfies either way. All four were rewritten before being trusted.
- **Control needle (§11.4.201(7)(b)):** re-injecting the exact original TMX-078 collision into a DB copy makes the validator FAIL naming TMX-078, while the tracked DB reports 0 findings across 3 iterations — the live-corpus zero is proven not blind.

Validation

Every figure below is **pinned to the revision it was measured at.** The full `verify.sh` sweep HAS now completed on the current tree — its result, its enumerated skips, and its honest boundary are recorded below the table.

Measurement	Revision	Result
`go vet ./...`	`ec25327`	clean
`go test -count=1 ./...`	`ec25327`	ok
Test 51	`ec25327`	`PASS=5 FAIL=0 SKIP=0` on 3/3 iteration
	`ec25327`	

Measurement	Revision	Result
`db-to-md` round-trip byte-identity		3/3, `hash=9d91e1a1e871150aec5fa8eebea9a0`
Sentinel rows (`code_ordinal <= 0`)	`8dad4e3`	17 → 3
Sentinel rows (category `Z`)	`8dad4e3`	4 → 0
Cross-tracker duplicates	`8dad4e3`	5 → 0
Sync idempotency	`8dad4e3`	`unchanged=92`, zero rebinds
`validate` findings	`8dad4e3`	0
`verify.sh` sweep	`1690789`	`exit 0`, GREEN, every summary `FAIL=` (hardware/topology, pre-existing)
`verify.sh` on the committable tree	**current (`2c8187a` + uncommitted)**	**COMPLETED 2026-09-02T06:09Z — `PA` `FAIL=0 SKIP=11`, GREEN** (`verify_final` earlier 05:42Z sweep predates later tr and does NOT validate this tree)
`CM-TRACKER-STRUCTURAL-INTEGRITY` (inside that sweep)	current	PASS — `6 tracker-structural check(`
`CM-REVIEW-ROUND-RECORD-WIRED` (inside that sweep)	current	PASS — `PASS=23 FAIL=0`
`CM-HEADING-GRAMMAR` (inside that sweep)	current	PASS — `no code-less tracker headin`
`heading_grammar_gate.sh` standalone	current	`PASS=2 FAIL=0 SKIP=0`, `rc=0`; Issues.mc coded, Fixed.md 86/86 coded
`heading_grammar_gate_test.sh` (self-test)	current	`PASS=9 FAIL=0` — incl. 2 §1.1 mutation RED on `HEAD:` artifacts
Full meta-test mutation sweep	**current**	**NOT RUN against this tree**
§11.4.185 manual QA	**current**	**NOT PERFORMED**

The completed sweep, in full (§11.4.6 honest boundary). It closed `GREEN: tmux binary verified — safe to PATH-export.` on HEAD `2c8187a` plus the uncommitted engine / gate / corpus work described above. The **11 SKIPs are honest topology / capability skips (§11.4.3), not passes**, and are enumerated rather than aggregated:

- `08_oom_score_adj`, `12_memory_pressure_under_cap`, `13_tasksmax_stress`, `14_concurrent_oom_independence` — destructive; require `TMX_TEST_DESTRUCTIVE=1` on a dedicated test host.

- `44_clipboard_copy_out_physical`, `45_multiline_copy_physical`, `46_paste_in_physical`, `48_modifier_drag_override` — require a GUI clipboard, absent on this headless host.
- `22_codegraph_mcp_wired`, `32_ssh_dispatch_remote_nezha`, `62_distribution_orchestrator` — absent topology.

The sweep is a verdict on the automated suite for this tree and nothing more: it does NOT discharge §11.4.185 manual QA (still owed), it does NOT cover the 11 skipped topologies, and it does NOT stand in for the on-device / dual-host suites those skips name. **COMPLETED 2026-09-02T06:09Z** (`verify_final.log`, 1177 lines). An EARLIER sweep completed 05:42Z (`verify_full.log`); the tree was then edited further (`.gitignore`, `README.md`, `CONTINUATION.md`, `Issues.md`, `Fixed.md`, 14 regenerated exports), so the 05:42Z run does NOT validate this tree and the 06:09Z run does. `verify.sh` does read `README.md` and `CONTINUATION.md`, which is why the re-run was required rather than optional. Every edit after 06:09Z is documentation-only and is re-validated by the §11.4.40 full retest that gates the tag. `grep -rln 'CHANGELOG\|docs/changelogs' scripts/` matches two comment lines only (`59_oomd_preference_avoid.sh:64`, `61_no_libtinfo_version_warning.sh:45`).

The 3 surviving `code_ordinal <= 0` rows are **not** sentinels: they are TMX-038 / TMX-039 / TMX-041, the genuine `### A0.` / `### B0` / `### C0` blocks at `Fixed.md:2585 / 2652 / 2718`, whose ordinal legitimately IS zero. The 14 rows that healed (TMX-057..066, TMX-072..075) were the true sentinels; the 14 newly-tracked blocks (TMX-079..092) are a DIFFERENT set of 14.

Accounting for 94 DB rows vs 92 parsed blocks (reproducible)

94 rows = 91 distinct block identities `(location, CAT, ORD)` + 2 rows sharing a block CODE with another row (TMX-001/TMX-054 at `Fixed B3`; TMX-071/TMX-062 at `Fixed A52`) + 1 row whose claimed identity has no block at all (TMX-050 at `Fixed F1`). 92 parsed = 91 distinct + 1 duplicated block CODE. All three extra rows are `Obsolete`, exactly what the audit's Obsolete guard skips, which is why `validate` correctly reports 0 findings.

Instruments discarded en route (§11.4.201)

Three measurement attempts produced confident wrong numbers **without crashing**, and each is recorded because the failure shape is the lesson.

Reported	Why it was wrong
"62 rows missing from markdown"	joined rows to blocks on the `TMX-ID` literal, but 60 of 85 `Fixed.md` blocks declare no such line — it measured the known missing-id gap, not row absence (§11.4.201(9): an id literal read as a row key)
"79 rows unbound by heading hash"	reimplemented `computeHeadingHash` over the RAW heading remainder; `parser.go:201` hashes `cleanTitle`, with the trailing `STATUS` hint already stripped by `trailingStatusRE`. With the strip, unbound = 2
"the row gap is the two refused headings"	no DB row corresponds to either heading, so they contribute ZERO rows — the same §11.4.201(9) field-identity class as the first row

The first two were caught and discarded during the investigation. The third was **published** and is RETRACTED in `2c8187a` and in the closure evidence: its conclusions (gap = 2, no live collision, `validate` 0 findings) survive, but they were reached via the wrong rows, and the arithmetic above is the reproducible replacement.

Honest regression in one number

The identity-audit **unlocated count ROSE 55 → 57**. This is explained, not hidden: 14 newly-visible blocks entered the SSoT, and the audit can only LOCATE a block that declares a `TMX-ID` line, which 60 of 85 `Fixed.md` blocks (and 3 of the 8 `Issues.md` blocks censused at the time) do not. Those blocks assert no ownership and are skipped by design — counted and reported out loud, never silently zeroed (§11.4.201(6)).

Review trail (§11.4.134 iterate-to-GO)

The heading-regex batch went round 1 NO-GO (1 BLOCKING — the author's own corpus-removal loop over-deleted the `## H.` header and its §11.4.114 preamble; restored verbatim from `git show HEAD:Issues.md`) → round 2 NO-GO (3 IMPORTANT

- 3 MINOR, and it VERIFIED the round-1 BLOCKING closed) → round 3 NO-GO (1 IMPORTANT: the `itemContentEqual` identity gap, found in the seam the author had asked the reviewer to attack) → round 4 **GO** → delta review of the two filed items **GO** → delta review of the tracker/evidence text NO-GO on 3 MINOR documentation-accuracy findings, all three remediated in `1690789` and `2c8187a`.

Round-numbering honest boundary (§11.4.6): no per-round artifact was written to disk, so the boundary could not be reconstructed from local state. The numbering is the reviewer's record, corroborated decisively by its own argument that a mutation which BECAME a round-2 finding cannot have survived round 1.

The review of the CURRENT batch is a separate, still-open loop — see below.

Known-open — explicitly NOT claimed fixed (§11.4.6)

1. **The independent review of the current batch returned NO-GO** — the LATEST round (2026-09-02, §11.4.209 substrate) reported 0 BLOCKING, 4 IMPORTANT, 1 MINOR, all documentation/hygiene, code layer clean; the PRIOR round reported 3 BLOCKING, 6 IMPORTANT, 5 MINOR — and has NOT returned a clean GO. §11.4.134 requires iterating to ZERO findings AND ZERO warnings. All three BLOCKING findings were confirmed by independent re-measurement before acceptance and are fixed in this release (the two unwired gates; a `CONTINUATION.md` freshness stamp bumped with zero content change; a closure-evidence citation this same batch falsified).
2. **The four reviewer-authored mutations that SURVIVED the prior round are REMEDIATED** (`cmd/workable-items/sourceless_signals_test.go`). The LATEST round ran 9 reviewer-authored mutations and measured 8 CAUGHT, 1 SURVIVED — and that survivor is the pre-declared, re-measured honest boundary the author documented in that same file (the `owners[it.ATMID]` half of

`blockClaimedBy` is defensive redundancy, not a load-bearing signal), NOT an escape. The §11.4.194(6)(d) pattern that produced the prior round's four survivors is recorded as the reason reviewer-authored mutations are mandatory, not as a closed-by-assertion item.
3. **`close.go`'s collision guard REFUSES a bad closure; it does not RENUMBER one.** The guard stops a NEW two-items-one-block state from being written and names the concrete recovery, but the renumber is manual — an automatic free-code allocation on close is not implemented and is not claimed. Nor does it repair an EXISTING stale pointer: TMX-050 still names `F1`, a block existing in NEITHER tracker. That repair is OWED.
4. **The missing `TMX-ID` backfill is OWED**, which is why **57 of 95 items remain unlocatable by the identity audit**. That number is reported out loud by `validate` rather than counted clean, and it is not claimed to be shrinking this cycle.
5. **The full meta-test mutation sweep has NOT been run against this tree.** Its outcome is not known and is not claimed. The 19 individually-pinned §1.1 mutations recorded in the closure evidence are not a substitute for the whole-suite mutation sweep. *(Two items formerly listed here are now CLOSED and moved to "Changed" / "Validation": the three code-less ### headings are repaired and the class is covered by the wired `CM-HEADING-GRAMMAR` gate; and the full `verify.sh` sweep on this tree COMPLETED GREEN — `PASS=79 FAIL=0 SKIP=11`.)*
6. **§11.4.185 manual QA has NOT been performed**, and the tag has NOT been cut. Both remain preconditions of the release, alongside the §11.4.40 full retest.
7. **This cycle closes NONE of the v1.0.44 known-open items.** Test 68 C6/C7 (TMX-080 / TMX-081 family), TMX-090 (`17_scrollback_copy_mode.sh` T3 live `WheelUpPane` binding), test 71 C4 on the `3.7b` artifact, and the `Fixed.md` §L placeholder closure SHAs all remain open exactly as recorded in the v1.0.44 release entry.
8. **§11.4.44 revision headers are absent from `Issues.md`, `Fixed.md` and `CONTINUATION.md`** (TMX-101, Queued). Measured 2026-09-02: Issues.md 0/0, Fixed.md 0/0, CONTINUATION.md 0 Revision / 1 Last-updated; control needle `docs/workable-items/Status.md` = 1, so the detecting grep is not blind. Repo-wide 60 of 86 tracked `.md` carry the field. PRE-EXISTING — not introduced by this batch, and NOT fixed in it.
9. **`Issues_Summary.md` and `Fixed_Summary.md` do not exist anywhere in the repository** (TMX-102, Queued), though §11.4.12 and §11.4.53 mandate both and `scripts/tests/51_workable_items_db_integrity.sh` references them. `git ls-files`

`| grep -iE 'Issues_Summary|Fixed_Summary'` is empty while the same search finds `Issues.md` and `Fixed.md` (control needle). PRE-EXISTING — not introduced by this batch, and NOT fixed in it.

10. **`sync db-to-md` WRITES the git-tracked SSoT DB on every run**, including a pure verification round-trip that renders into a throwaway `--out-dir` (TMX-103, Queued). `cmd/workable-items/sync_db_to_md.go:131-134` rewrites `meta.last_sync_direction` / `meta.last_sync_timestamp` unconditionally, so the documented byte-identity verification procedure dirties the tracked `docs/workable_items.db` and cannot be restored byte-exactly. Scope proof: successive runs on copies differ ONLY in those two meta rows (full logical table dump); `validate` is byte-neutral. PRE-EXISTING at `HEAD:cmd/workable-items/sync_db_to_md.go` — this batch newly enshrines the round-trip as the verification mechanism, and does NOT fix the side effect.

Discovery direction (§11.4.238)

None of this cycle's defects was operator-reported. The heading-form gap surfaced from a tracker-integrity investigation; six more came out of fixing it; TMX-093 / TMX-094 / TMX-095 and the two audit defects came from a five-stream root-cause fan-out (§11.4.230(C)); and the three BLOCKING findings — including two gates that enforced nothing — came from the independent review. That is the intended direction of discovery, and it is also why the review loop is not yet closed.

Files changed

- **Engine:** `cmd/workable-items/{parser.go, reconcile.go, db.go, sync_md_to_db.go, sync_db_to_md.go, add.go, close.go, validate_identity.go, main.go}`.
- **Tests:** `cmd/workable-items/{heading_forms, duplicate_block, upsert_identity, guard_coverage, validate_identity, validate_identity_duplicate, add_hash_convention, add_category_validation, db_to_md_add_rows, sourceless_signals, close_identity, parser_greedybind, reconcile_identity}_test.go`; `scripts/tests/51_workable_items_db_integrity.sh`.
- **Gates:** `scripts/verify.sh`; `scripts/testing/tracker_structural_integrity{,_test}.sh`; `scripts/review/review_round_record{,_test}.sh`; `scripts/testing/heading_grammar_gate{,_test}.sh`.

- **Corpus + SSoT:** `Issues.md`, `Fixed.md`, `docs/workable_items.db`.
- **Docs:** `CHANGELOG.md`, `CONTINUATION.md`, `README.md`, `docs/scripts/{tracker_structural_integrity,review_round_record,heading_grammar_gate}.md`, `docs/qa/2026-09-01-heading-regex-widening/closure-evidence.md`, this file.
- **Submodule:** `constitution` pointer `b9096ac` → `f5876a3`.
- **Pending at the time of writing:** `VERSION` (1.0.45 / versionCode 46) is bumped when the tag is cut, not here.

Sources verified 2026-09-01

NO external solution found — original work (§11.4.8). Every defect above is internal to this project's own tracker engine, gate wiring, and corpus; no third-party documentation, API, or upstream behaviour was consulted or relied on, so there is no external source to cross-reference under §11.4.99.